

Lab 1 — Building Your First Interactive Dashboard in Power BI

Cards • Tables • Matrix • Slicers • Relationships (Star Schema)

1. Lab Overview & Learning Outcomes

In this lab you will take seven raw data files — one large **fact table** and six **dimension tables** — and turn them into a clean, connected data model and an interactive dashboard. You will not need any prior Power BI experience; every click is written out. Work through the parts in order, because each part builds on the previous one.

By the end of this lab you will be able to:

- Import CSV and Excel files into Power BI Desktop.
- Perform simple data cleaning in Power Query (fix a bad header row, convert text to numbers).
- Create and verify relationships between a fact table and its dimensions (a star schema).
- Build KPI Cards that summarise the business at a glance.
- Present detail with a Table and a Matrix visual.
- Add Slicers and understand how visuals cross-filter one another.

2. Before You Begin

- **Software:** Microsoft Power BI Desktop (free). Open it and start with a blank report.

3. The Dataset — A Star Schema

The data describes bicycle and accessory sales for a company called AdventureWorks. It is organised as a **star schema**: one central table of measurable events (sales), surrounded by descriptive tables (who, what, when, where). The central table is the **fact table**; the surrounding tables are **dimension tables**.

File	Role	Key columns & notes
FactSales.csv	Fact	121,000+ rows. Keys: ProductKey, CustomerKey, ResellerKey, SalesTerritoryKey, OrderDateKey, SalesOrderLineKey. Measures: Order Quantity, Sales Amount, Total Product Cost.
DimProducts.xlsx	Dimension	ProductKey, Product, Category, Subcategory, Color, List Price.
DimCustomers.csv	Dimension	CustomerKey, Customer, City, Country-Region. Note: has an extra junk row at the top you will fix.
DimReseller.csv	Dimension	ResellerKey, Reseller, Business Type, Country-Region.
DimSalesTerritory.csv	Dimension	SalesTerritoryKey, Region, Country, Group.
DimSalesOrder.csv	Dimension	SalesOrderLineKey, Sales Order, Channel (Reseller / Internet).
DimDate.csv	Dimension	DateKey, Date, Fiscal Year, Fiscal Quarter, Month.

Part A — Load the Data

1. Open Power BI Desktop and close the start screen to reach a blank report.
2. On the **Home** ribbon click **Get data** ▼ → **Text/CSV**. Select `FactSales.csv` and click **Open**.
3. A preview window appears. For now click **Load**. The table appears in the **Data pane** on the right.
4. Repeat **Get data** → **Text/CSV** for these five files, clicking **Load** each time: `DimCustomers.csv`, `DimReseller.csv`, `DimSalesTerritory.csv`, `DimSalesOrder.csv`, `DimDate.csv`.
5. For the products file use **Get data** → **Excel workbook**, select `DimProducts.xlsx`, tick the **DimProducts** sheet in the Navigator, then **Load**.
6. Check the Data pane now lists all **seven** tables.

Part B — Quick Data Cleaning

Two problems must be fixed before the model will work. Open the editor: **Home** ribbon → **Transform data**. This opens the **Power Query Editor** (a separate window). The tables are listed on the left under **Queries**.

B1. Fix the junk header row in DimCustomers

The customers file has an extra line at the very top, so the column names are wrong.

1. Click the **DimCustomers** query on the left.
2. On the **Transform** ribbon click **Use Headers as First Row** ▼ → **Use Headers as First Row** (this pushes the bad names down into the data).
3. On the **Home** ribbon click **Remove Rows** ▼ → **Remove Top Rows**, type 1, click **OK**.
4. On the **Home** ribbon click **Use First Row as Headers**. The columns should now read `CustomerKey`, `Customer ID`, `Customer`, `City`....

B2. Convert money columns from text to numbers in FactSales

The amounts were imported as text (e.g. \$2,024.99) because of the \$ sign and commas, so Power BI cannot add them up yet.

1. Click the **FactSales** query on the left.
2. Hold **Ctrl** and click the headers of these five columns to select them all: `Unit Price`, `Extended Amount`, `Product Standard Cost`, `Total Product Cost`, `Sales Amount`.
3. Right-click any selected header → **Change Type** → **Using Locale....**
4. In the dialog set **Data Type** = **Fixed decimal number** (or **Decimal Number**) and **Locale** = **English (United States)**, then **OK**. The \$ and commas disappear and the values become real numbers.
5. Confirm `Order Quantity` is **Whole Number** (its icon is 123). If not, change its type to **Whole Number**.

B3. Apply your changes

1. On the **Home** ribbon click **Close & Apply**. Power BI returns to the report and reloads the cleaned data.

Part C — Build & Verify Relationships

Switch to **Model view** using the third icon on the far-left rail (it looks like connected boxes). You will see your seven tables. Power BI may have auto-created some relationship lines already — you will check them and add any that are missing.

To create a relationship, **drag the key field from a dimension onto the matching key in FactSales**. Make sure every one of these six links exists:

From (dimension)	To (fact table)	Type
DimProducts[ProductKey]	FactSales[ProductKey]	1 → *
DimCustomers[CustomerKey]	FactSales[CustomerKey]	1 → *
DimReseller[ResellerKey]	FactSales[ResellerKey]	1 → *
DimSalesTerritory[SalesTerritoryKey]	FactSales[SalesTerritoryKey]	1 → *
DimSalesOrder[SalesOrderLineKey]	FactSales[SalesOrderLineKey]	1 → *
DimDate[DateKey]	FactSales[OrderDateKey]	1 → *

- The “**one**” (1) side is always the dimension; the “**many**” (*) side is always FactSales (a product appears on many sales lines).
- Double-click a relationship line to check **Cardinality = Many to one (*:1)** and **Cross-filter direction = Single**.
- The fact table has three date columns (OrderDateKey, DueDateKey, ShipDateKey). For this lab connect **only OrderDateKey** to DimDate. If Power BI made an extra date link, delete it (right-click → Delete) so just the OrderDateKey link remains solid/active.

Part D — KPI Cards

Switch back to **Report view** (top icon on the left rail). A **Card** shows a single headline number.

1. In the **Visualizations** pane click the **Card** icon. An empty card appears on the canvas.
2. With the card selected, drag FactSales[Sales Amount] into it. It should read a large total (it sums automatically).
3. Click a blank area, add a second Card, and drag in FactSales[Order Quantity] (Total Units).
4. Add a third Card with FactSales[Total Product Cost] (Total Cost).
5. Add a fourth Card, drag in DimSalesOrder[Sales Order], then click the small arrow next to the field in the Card and choose **Count (Distinct)** to get the number of orders.
6. Tidy each card: open the **Format** (paintbrush) pane and turn on **Title**, giving names like Total Sales, Total Units, Total Cost, # Orders.

Part E — Table & Matrix

E1. A Table of sales by category

1. Click a blank area, then the **Table** visual in the Visualizations pane.
2. Drag in DimProducts[Category] then FactSales[Sales Amount].
3. Click the **Sales Amount** column header inside the visual to sort from highest to lowest.

E2. A Matrix of category by year

4. Add a **Matrix** visual.
5. Set Rows = DimProducts[Category], Columns = DimDate[Fiscal Year], Values = FactSales[Sales Amount].
6. Notice the matrix shows a grid of sales with automatic row and column totals — this is the power of your relationships working together.

Part F — Slicers & Cross-Filtering

A **Slicer** is an on-canvas filter your audience can click. Add three.

1. Click a blank area → **Slicer** visual → drag in DimDate[Fiscal Year].
 2. Add a second Slicer with DimSalesTerritory[Group] (region).
 3. Add a third Slicer with DimProducts[Category].
 4. Click a value in a slicer (e.g. a fiscal year). Watch every card, the table and the matrix update instantly.
 5. Now click a single row in the **Matrix** or a cell — see the cards react. This is **cross-filtering**: selecting in one visual filters the others.
- **Tip:** hover a slicer and use the small ▼ to switch between **List** and **Dropdown** styles to save space.
 - **Tip:** Ctrl+click selects multiple slicer values; click again to clear.

Key Terms (Glossary)

Term	Meaning
Fact table	The central table holding measurable events and their numeric measures (e.g. sales).
Dimension table	A descriptive table that gives context — the who, what, when, where.
Star schema	A model with one fact table linked to several dimension tables, resembling a star.
Key	An ID column (...Key) used to link a fact row to its dimension row.
Relationship / Cardinality	The link between two tables and how many rows on each side match (usually one-to-many).
Measure vs Column	A column stores values per row; a measure is a calculation (e.g. a sum) evaluated on the fly.
Card	A visual that displays one summary number.
Slicer	An on-canvas filter the user can click to change what all visuals show.
Cross-filtering	Selecting in one visual automatically filters the other visuals.
Power Query	The Power BI tool used to import and clean data before it loads into the model.

Troubleshooting

Symptom	Fix
Sales Amount will not add up / shows text	Re-do Part B2: select the columns → Change Type → Using Locale → Currency, English (US).
A relationship line is dotted	Dotted means inactive. For this lab keep only the OrderDateKey→DateKey link active; delete extra date links.
Numbers turn into Error after changing type	The locale is wrong — set it to English (United States) so it understands \$ and commas.
A card is blank or wrong	Check the field's aggregation (Sum vs Count) and that the table is related to FactSales.
Customer columns are named Column1, Column2...	Redo Part B1 to fix the header row on DimCustomers.
Cannot find Model view	Use the icons on the far-left vertical rail: Report, Table, and Model views.